

Matrix Algebra

Part 1/3: Foundations and Matrix Operations

CSC 1105 Mathematics for Computer Science

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Module 3 at a Glance

Module 3: Matrix Algebra

This part introduces the fundamental language and rules of matrix algebra.

In the full module, we shall study:

- Matrix algebra rules
- Transposes, determinants and inverses
- Eigenvectors, kernels and matrix transformations
- Solving equations using matrices

Learning Outcomes

By the end of this lecture, you should be able to:

- 1 Define a matrix and identify its order
- 2 Recognize common types of matrices
- 3 Perform matrix addition, subtraction, and scalar multiplication
- 4 Check whether matrix multiplication is possible
- 5 Multiply matrices using the row-by-column rule
- 6 State key algebraic properties of matrix operations

Computer Science Connection

Matrices are useful in computer science because they can represent:

- Images as grids of pixel values
- Graphs and networks using adjacency matrices
- Transformations in graphics and robotics
- Data tables in machine learning
- Systems of equations in optimization and modelling

What is a Matrix?

Definition

A **matrix** is a rectangular array of numbers, symbols or expressions arranged in rows and columns.

$$A = \begin{bmatrix} 2 & -1 & 0 \\ 4 & 3 & 7 \end{bmatrix}$$

This matrix has:

- 2 rows
- 3 columns
- Order 2×3

Matrix Notation

A general $m \times n$ matrix is written as

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}.$$

- a_{ij} is the entry in row i and column j
- m is the number of rows
- n is the number of columns

Rows, Columns and Entries

Consider

$$A = \begin{bmatrix} 5 & 2 & -3 \\ 0 & 7 & 1 \\ 4 & -2 & 6 \end{bmatrix}.$$

Examples

$$a_{11} = 5, \quad a_{23} = 1, \quad a_{32} = -2.$$

Order

A is a 3×3 matrix.

Common Types of Matrices

Type	Description
Row matrix	Has one row
Column matrix	Has one column
Square matrix	Number of rows equals number of columns
Zero matrix	All entries are zero
Diagonal matrix	Nonzero entries appear only on the main diagonal
Identity matrix	Diagonal entries are 1 and other entries are 0

Examples of Matrix Types

Row Matrix

$$\begin{bmatrix} 2 & 5 & -1 \end{bmatrix}$$

Column Matrix

$$\begin{bmatrix} 3 \\ 0 \\ 8 \end{bmatrix}$$

Square Matrix

$$\begin{bmatrix} 1 & 4 \\ 2 & -3 \end{bmatrix}$$

Zero Matrix

$$0_{2 \times 3} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Identity Matrix

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Equality of Matrices

Two matrices are equal if:

- They have the same order
- Corresponding entries are equal

Example

If

$$\begin{bmatrix} x & 3 \\ 2 & y \end{bmatrix} = \begin{bmatrix} 5 & 3 \\ 2 & -1 \end{bmatrix},$$

then

$$x = 5, \quad y = -1.$$

Adding and Subtracting Matrices

Matrices can be added or subtracted only if they have the same order.

If $A = [a_{ij}]$ and $B = [b_{ij}]$ are both $m \times n$, then

$$A + B = [a_{ij} + b_{ij}], \quad A - B = [a_{ij} - b_{ij}].$$

Important

Matrix addition is done entry by entry.

Example: Matrix Addition

Let

$$A = \begin{bmatrix} 2 & -1 & 4 \\ 0 & 3 & 5 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 6 & -2 \\ 7 & 0 & 3 \end{bmatrix}.$$

Then

$$A + B = \begin{bmatrix} 2 + 1 & -1 + 6 & 4 + (-2) \\ 0 + 7 & 3 + 0 & 5 + 3 \end{bmatrix} = \begin{bmatrix} 3 & 5 & 2 \\ 7 & 3 & 8 \end{bmatrix}.$$

Scalar Multiplication

To multiply a matrix by a scalar, multiply every entry by the scalar.

If

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix},$$

then

$$kA = \begin{bmatrix} ka_{11} & ka_{12} \\ ka_{21} & ka_{22} \end{bmatrix}.$$

Example

$$3 \begin{bmatrix} 2 & -4 \\ 0 & 5 \end{bmatrix} = \begin{bmatrix} 6 & -12 \\ 0 & 15 \end{bmatrix}.$$

Combining Matrix Operations

Example

Given

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & 0 \\ -1 & 2 \end{bmatrix},$$

find $2A - B$.

$$2A = \begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix}$$

$$2A - B = \begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix} - \begin{bmatrix} 5 & 0 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} -3 & 4 \\ 7 & 6 \end{bmatrix}.$$

The Row-by-Column Rule

If

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad B = \begin{bmatrix} e & f \\ g & h \end{bmatrix},$$

then

$$AB = \begin{bmatrix} ae + bg & af + bh \\ ce + dg & cf + dh \end{bmatrix}.$$

Each entry of AB is found by multiplying a row of A by a column of B .

Matrix Multiplication: Example 1

Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 0 \\ 1 & 5 \end{bmatrix}.$$

Then

$$AB = \begin{bmatrix} 1(2) + 2(1) & 1(0) + 2(5) \\ 3(2) + 4(1) & 3(0) + 4(5) \end{bmatrix} = \begin{bmatrix} 4 & 10 \\ 10 & 20 \end{bmatrix}.$$

Matrix Multiplication: Example 2

Let

$$A = \begin{bmatrix} 2 & -1 & 3 \\ 0 & 4 & 5 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 \\ 3 & 0 \\ -2 & 1 \end{bmatrix}.$$

Since A is 2×3 and B is 3×2 , AB is defined and has order 2×2 .

$$AB = \begin{bmatrix} 2(1) + (-1)(3) + 3(-2) & 2(2) + (-1)(0) + 3(1) \\ 0(1) + 4(3) + 5(-2) & 0(2) + 4(0) + 5(1) \end{bmatrix} = \begin{bmatrix} -7 & 7 \\ 2 & 5 \end{bmatrix}.$$

Matrix Multiplication is Not Commutative

In general,

$$AB \neq BA.$$

Example

Let

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & 1 \\ 4 & 2 \end{bmatrix}.$$

Then

$$AB = \begin{bmatrix} 11 & 5 \\ 4 & 2 \end{bmatrix}, \quad BA = \begin{bmatrix} 3 & 7 \\ 4 & 10 \end{bmatrix}.$$

Useful Properties

When the matrix products are defined:

$$A + B = B + A$$

$$(AB)C = A(BC)$$

$$A(B + C) = AB + AC$$

$$(A + B)C = AC + BC$$

$$AI = IA = A$$

Note

The property $AB = BA$ is not generally true.

Transpose of a Matrix

The **transpose** of a matrix is obtained by interchanging rows and columns.

If

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix},$$

then

$$A^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}.$$

If A is $m \times n$, then A^T is $n \times m$.

Properties of Transpose

For suitable matrices A and B and scalar k :

$$(A^T)^T = A$$

$$(A + B)^T = A^T + B^T$$

$$(kA)^T = kA^T$$

$$(AB)^T = B^T A^T$$

Important

The order reverses in the transpose of a product.

Symmetric Matrices

A square matrix A is called **symmetric** if

$$A^T = A.$$

Example

$$A = \begin{bmatrix} 2 & -1 & 4 \\ -1 & 0 & 3 \\ 4 & 3 & 5 \end{bmatrix}$$

Since $A^T = A$, the matrix is symmetric.

Matrices and Systems of Equations

A system of linear equations can be written in matrix form.

$$\begin{aligned}2x + y &= 5 \\ 3x - 4y &= -2\end{aligned}$$

becomes

$$\begin{bmatrix} 2 & 1 \\ 3 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ -2 \end{bmatrix}.$$

That is,

$$A\mathbf{x} = \mathbf{b}.$$

Exercise

Let

$$A = \begin{bmatrix} 1 & 0 & 2 \\ -1 & 3 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 1 \\ 0 & 4 \\ 5 & -2 \end{bmatrix}.$$

- 1 State the order of A and B .
- 2 Is AB defined? If yes, state its order.
- 3 Is BA defined? If yes, state its order.
- 4 Compute AB .

Questions?